

A Robust Fetal ECG Detection Method for Abdominal Recordings

Reproducing and evaluating Martens et al. (2007) on the CinC 2013 dataset

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Biomedical Signal Processing Project

Introduction

Clinical Context

- Fetal cardiac activity is a key indicator of fetal *well-being*
- Routine Doppler ultrasound only provides the **Fetal heart rate (FHR)**, not morphology
- **Fetal ECG (FECG)** carries *PR/QRS/ST morphology*, which are directly linked to *fetal oxygenation*
- *Abdominal FECG* is a non-invasive method usable throughout pregnancy *and* labour (vs. *invasive scalp electrode, labour-only*)

Challenge

- *FECG amplitude: tiny* (tens of μV)
- *Buried under: baseline wander, power-line interference, EMG/motion artifacts*
- *Above all: the **mother's ECG (MECG)** is an order of magnitude *larger*, same electrodes*
- *Goal: strip away everything that isn't the fetal heartbeat*

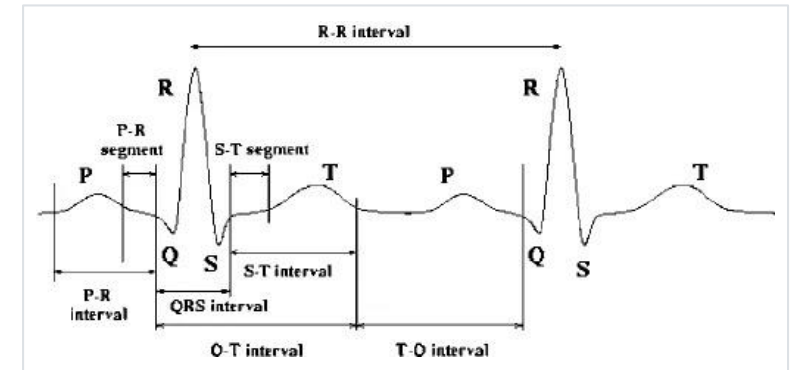


Figure 0. nomenclature of the ECG.

Two paradigms

Blind (ICA / BSS)

- No prior knowledge assumed
- Relies purely on statistical independence of sources
- As many independent sources as channels required

vs.

Non-blind (Sequential Analysis)

- Exploits a *priori* knowledge of each interferer
- Uses periodicity, morphology, frequency content
- Thesis: more robust, especially at low **SNR**

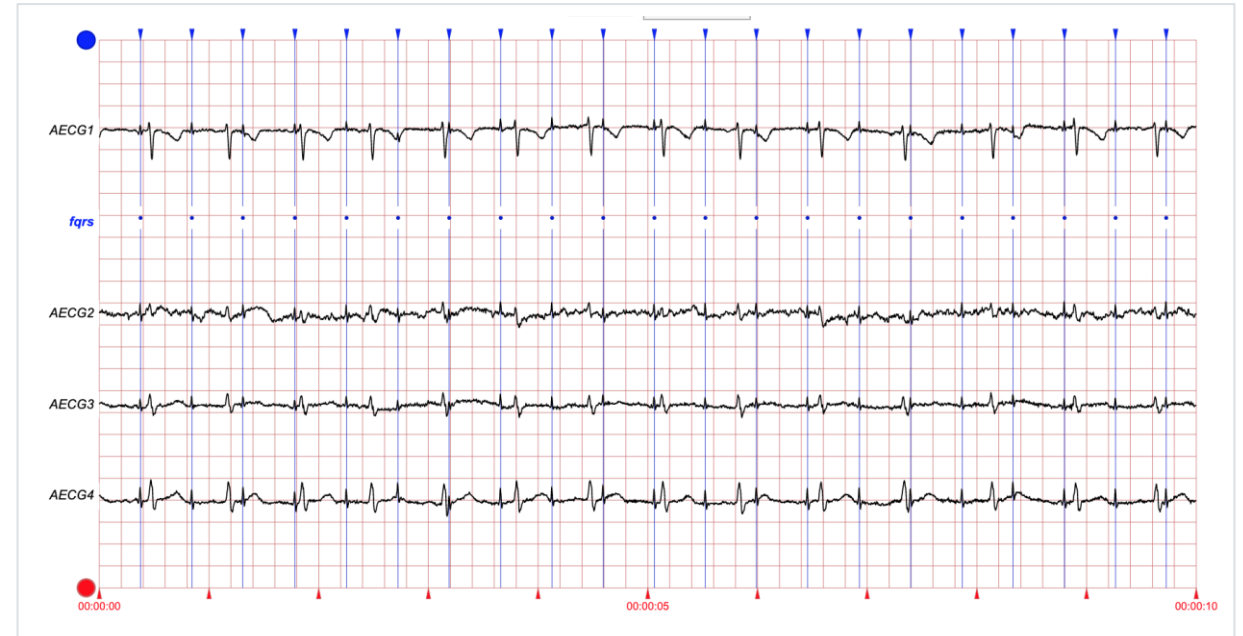
Pipeline overview



Note: signals were originally sampled at **1000 Hz** → upsampled to **2000 Hz** for MECG cancellation precision

Dataset

- *Sourced from:* **The 2013 PhysioNet/CinC Challenge**
- *Data structure:* 1 minute recordings, 4 non-invasive addominal channels
- *Sampling frequency:* 1000 Hz
- *GT annotations:* spatial locations of each fetal QRS complex are included



Stage 1–2: BW removal + PLI canceller

- *Baseline wander removal*: linear-phase FIR high-pass, 3 Hz cutoff, Hamming window
- *Power-line interference canceller*: adaptive PLL-style canceller with a QRS-blocking mask
- *Results*: baseline flattened and stabilized. Removal of the 50 Hz interference and its harmonics up to 200 Hz.

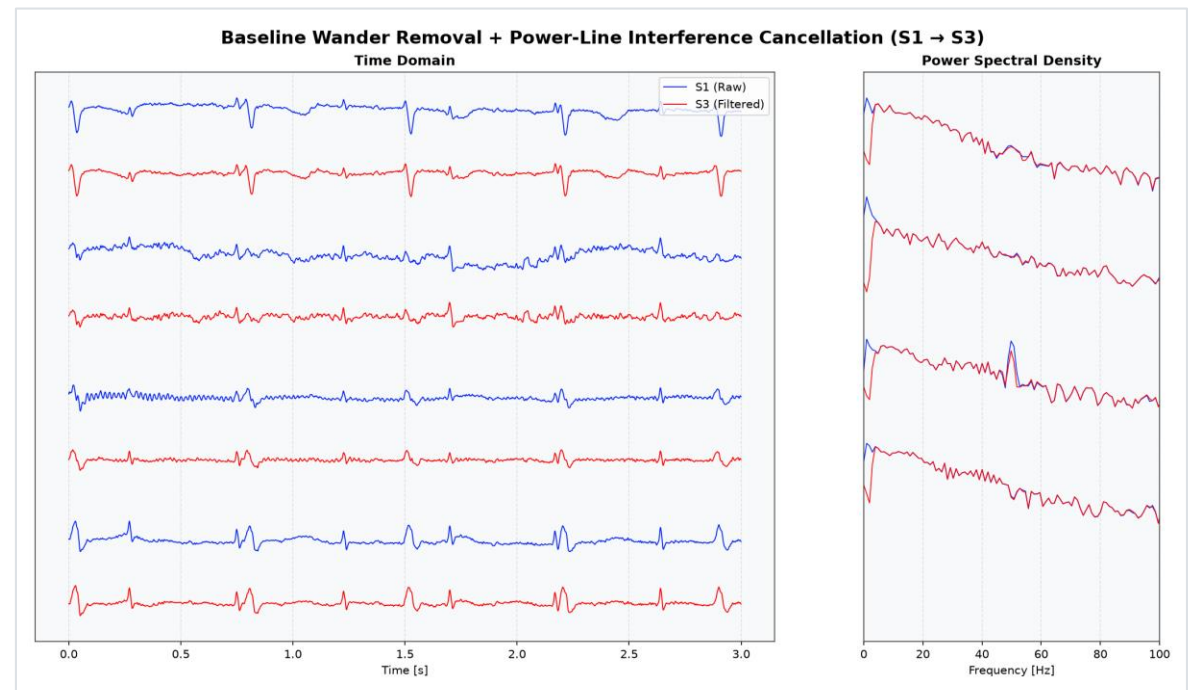
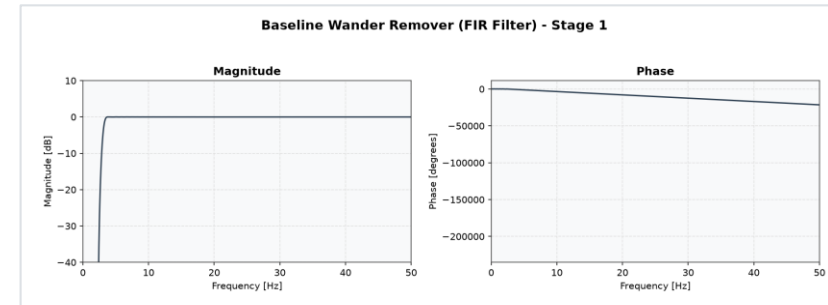


Figure 1. baseline wander + power-line interference

Stage 3–4: maternal QRS detection & MECG cancellation

- *Maternal QRS detection*: PCA channel combination (maximises SNR) + cross-correlation matched filter
- *MECG cancellation*: average P-wave/QRS/T-wave templates, each **scaled independently** via least squares
- *Numerical Stability*: **ridge Tikhonov regularisation** applied to the least squares fit

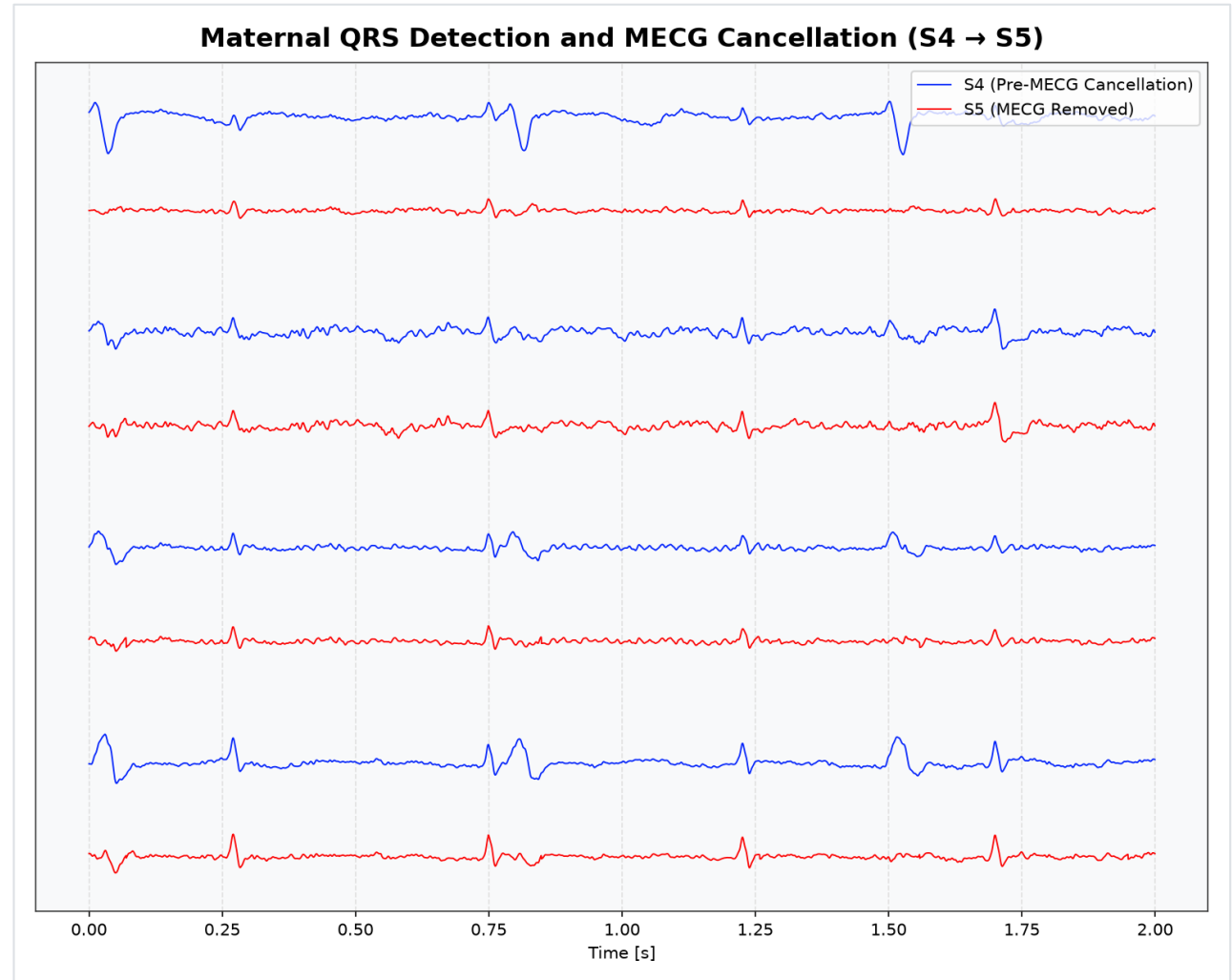


Figure 2. maternal QRS detection & MECG cancellation

Stage 5: fetal QRS detection & FECG extraction

- *Signal enhancement*: PCA applied to the MEGC free signal
- *Candidate peaks*: squared signal **energy** **thresholding**
- *Refinement*: cross-correlation matched filter
- *Final step*: synchronous averaging of ~150 beats to achieve an SNR boost of VN

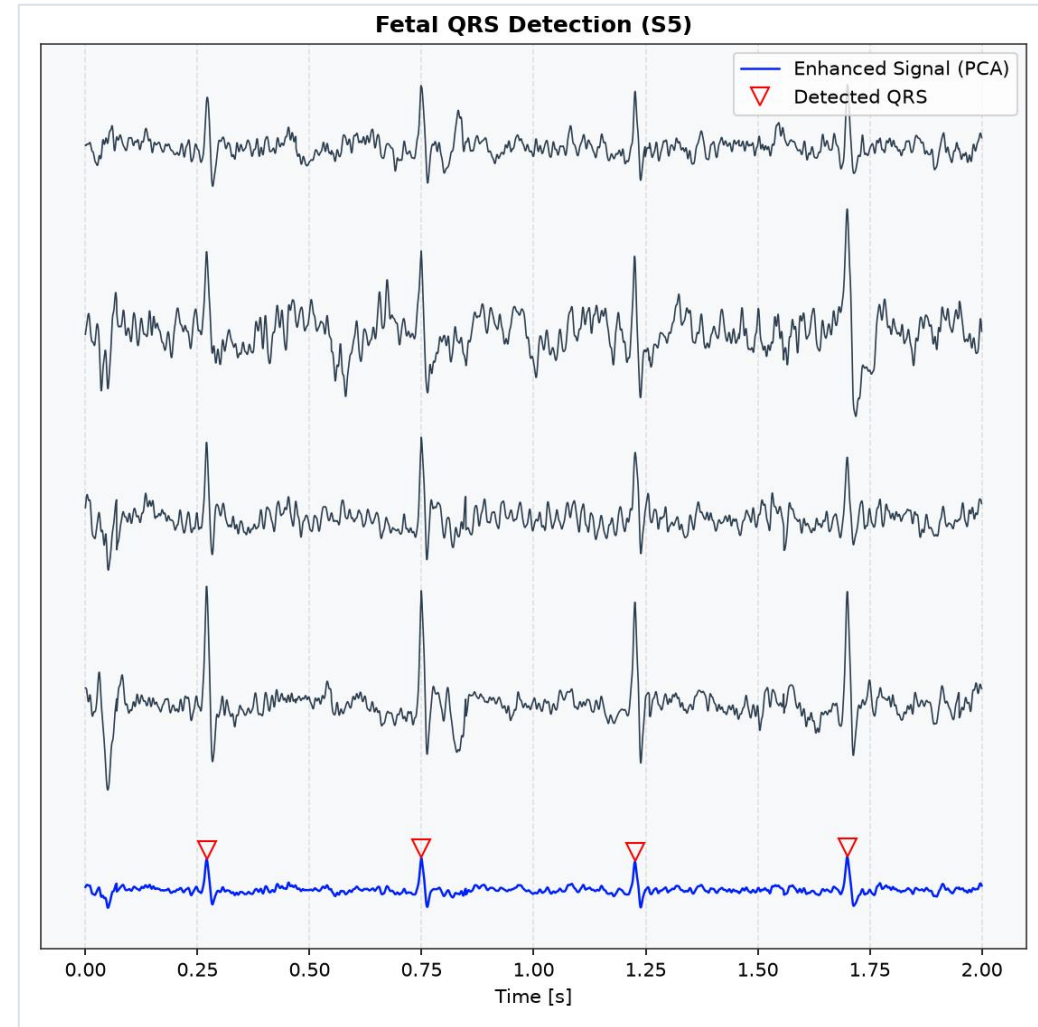


Figure 3. fetal QRS detection & FECG extraction

Full pipeline at a glance

- Raw → cleaned → MEG removed → clean averaged fetal beat

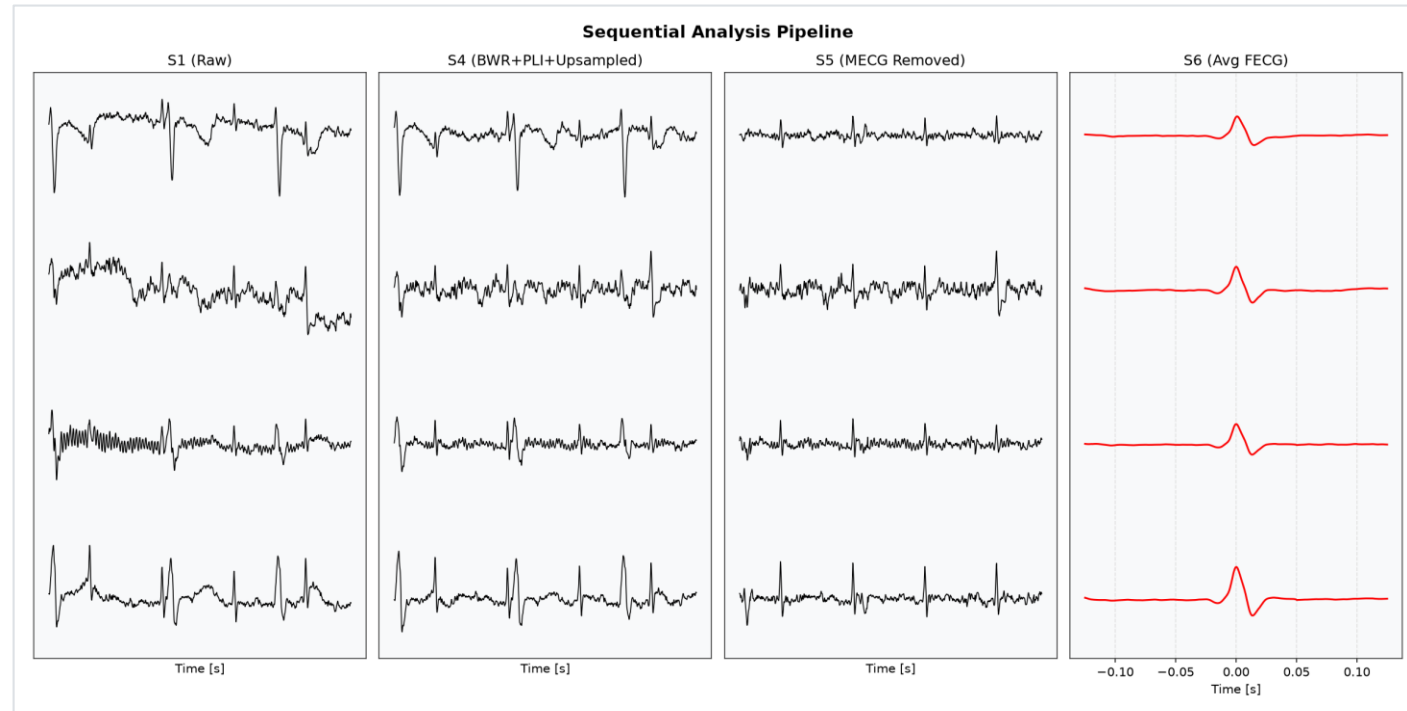


Figure 4. Full pipeline at a glance

Qualitative validation against ground truth

- Detected fetal QRS peaks (red) vs. ground truth annotations (blue), ± 50 ms tolerance
- 10 second, 4 channel strip
- Averaged fetal morphology across 124 ground truth matched beats

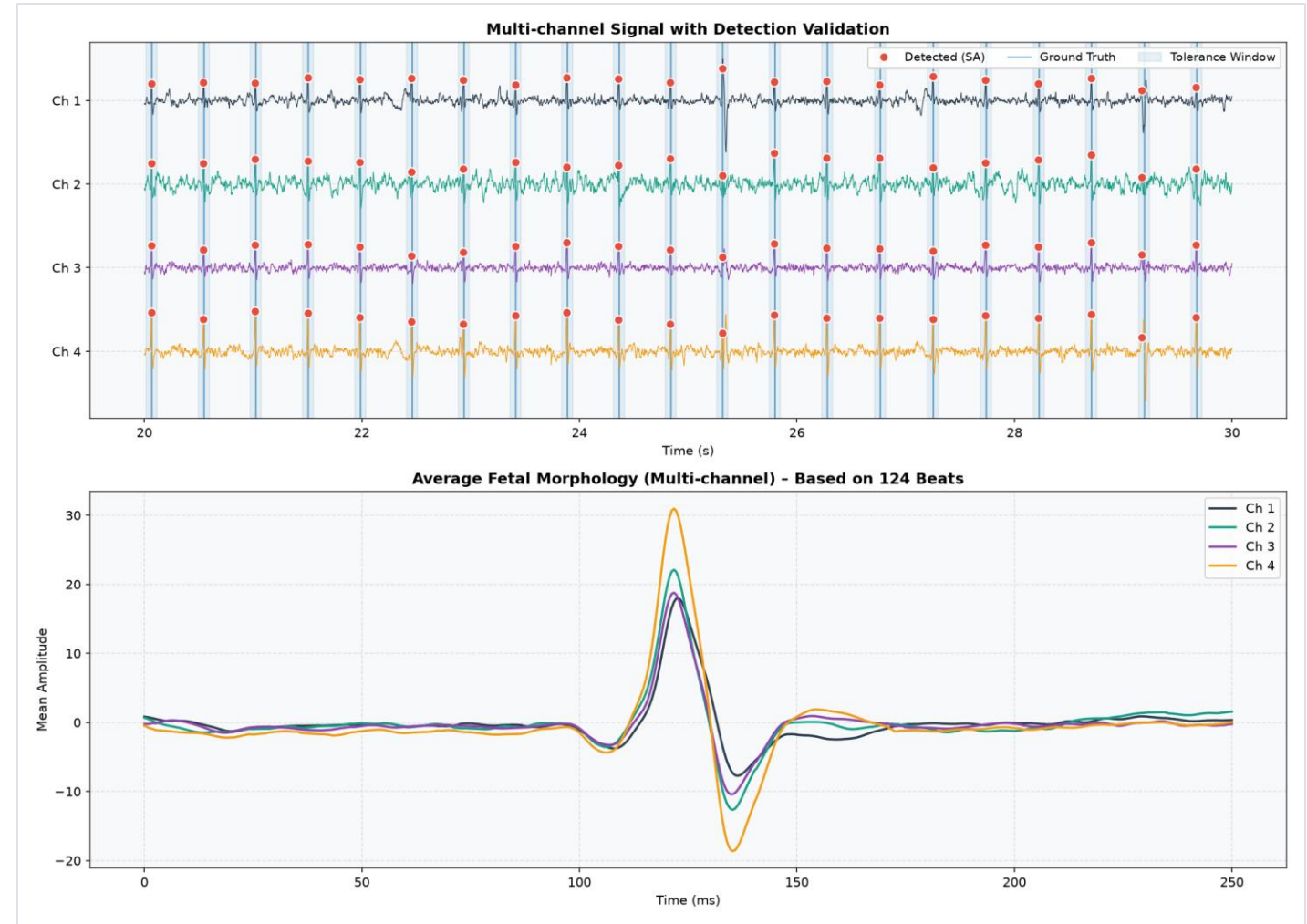


Figure 5. Qualitative validation against ground truth

Baseline comparison: the ICA pipeline

- FastICA run on the same preprocessed data (baseline wander + PLI removed) as the *"blind"* alternative
- Best independent component auto selected by FHR detection yield

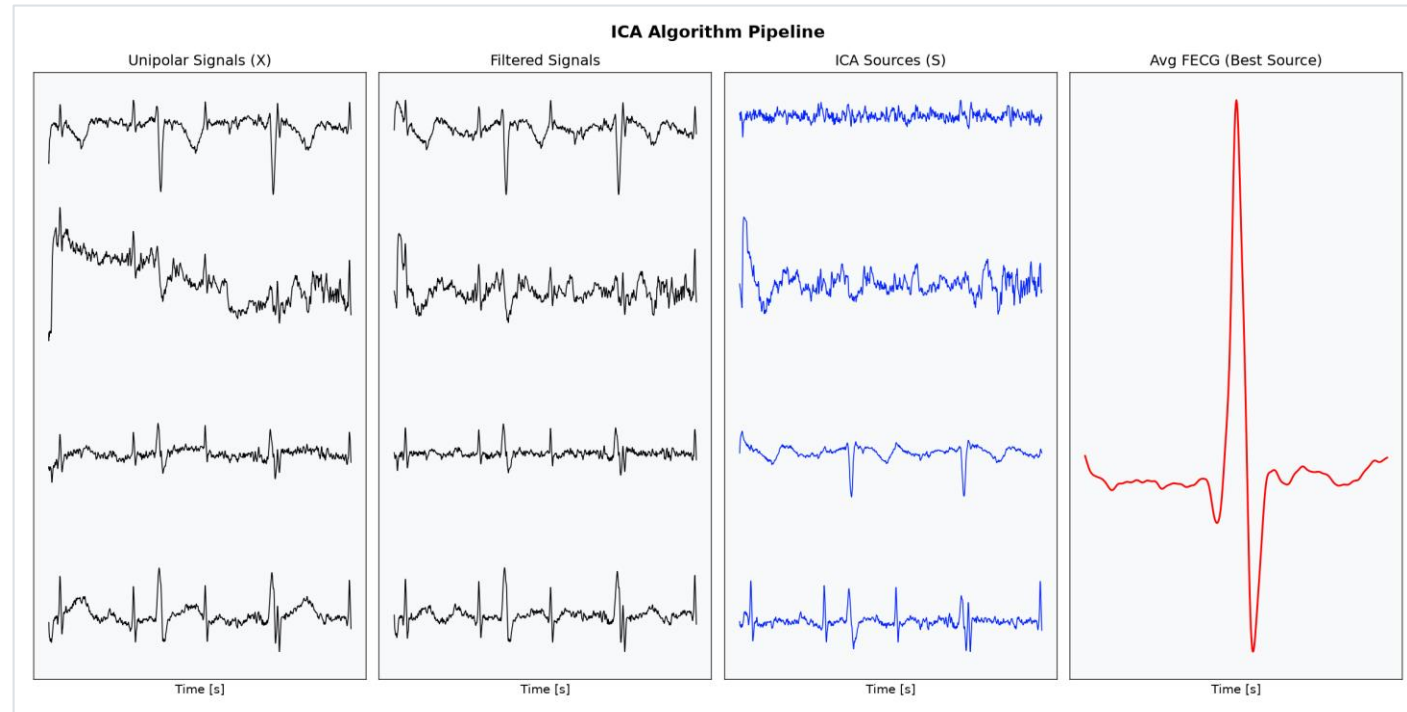


Figure 6. Baseline comparison: the ICA pipeline

Evaluation metrics

Success

coverage $\geq 60\%$ of the recording

coverage $\geq 60\%$

Reliability

outlier: deviates >10 bpm from its 10 s block's median

$$R = 1 - \frac{n_{\text{outliers}}}{n_{\text{total}}}$$

SNR / SIR (dB, per channel j)

FECG, MECG, and noise power taken from S6, S4–S5, and the residual

$$\text{SNR}_j = 10 \log_{10} \left(\frac{P_F}{P_N} \right)$$

$$\text{SIR}_j = 10 \log_{10} \left(\frac{P_F}{P_M} \right)$$

Precision / Recall / F1

matched to ground truth at ± 50 ms tolerance

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

FHR-value MAE / RMSE (bpm)

estimated vs. ground truth FHR, common 1 s grid

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N | \text{FHR}_i^{\text{est}} - \text{FHR}_i^{\text{gt}} |$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\text{FHR}_i^{\text{est}} - \text{FHR}_i^{\text{gt}})^2}$$

Results I: success rate & reliability

- *FHR detection success rate: SA 92% vs. ICA 96% (23/25 vs. 24/25 records)*
- *Mean FHR reliability (feasible records only): SA 0.753 vs. ICA 0.684*
- *Box plots: SA's spread is tighter and more consistent — ICA is bimodal (several near perfect records, several collapsed as low as ~0.23)*

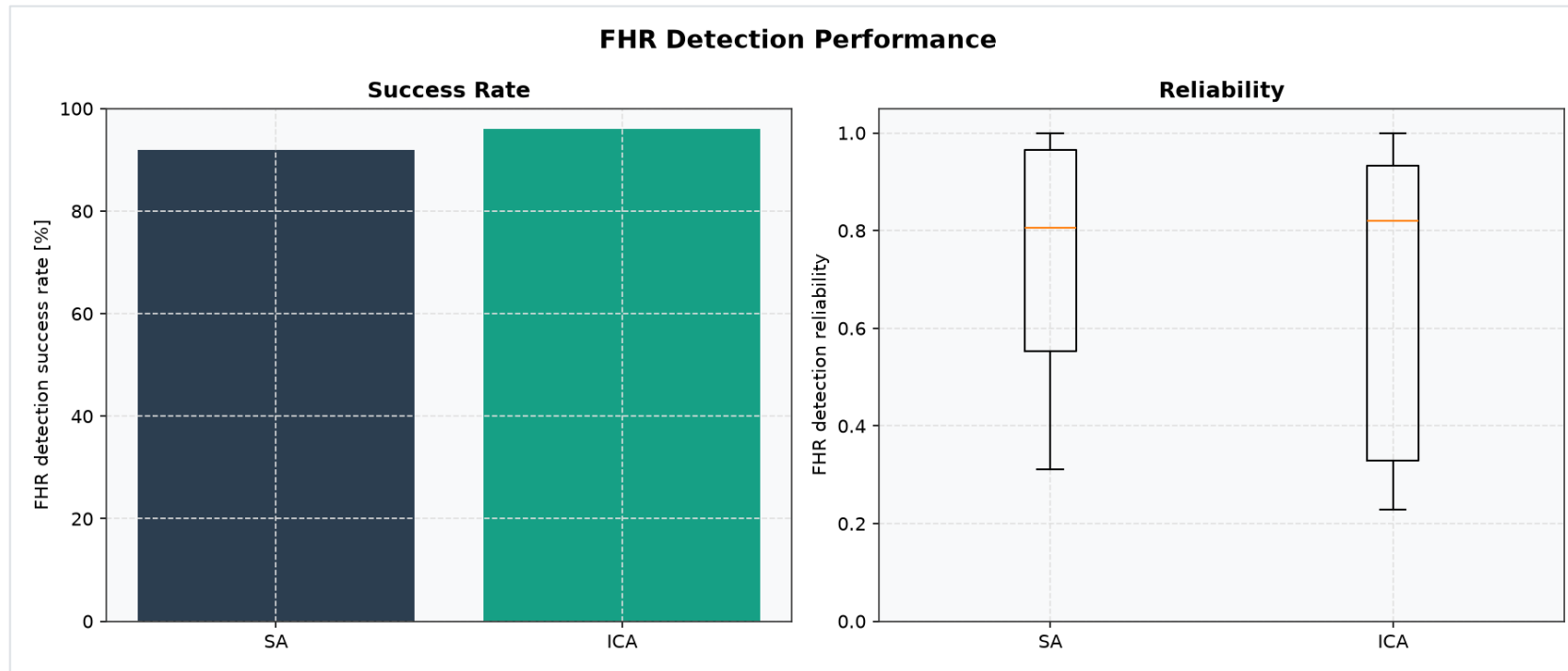


Figure 7. success rate & reliability

Results II: ground truth precision / recall / F1

- SA: **77.4%** precision / **67.1%** recall / **71.9%** F1
- ICA: **59.7%** precision / **53.1%** recall / **56.2%** F1
- Matched against ground truth annotations, ± 50 ms tolerance, all 25 records

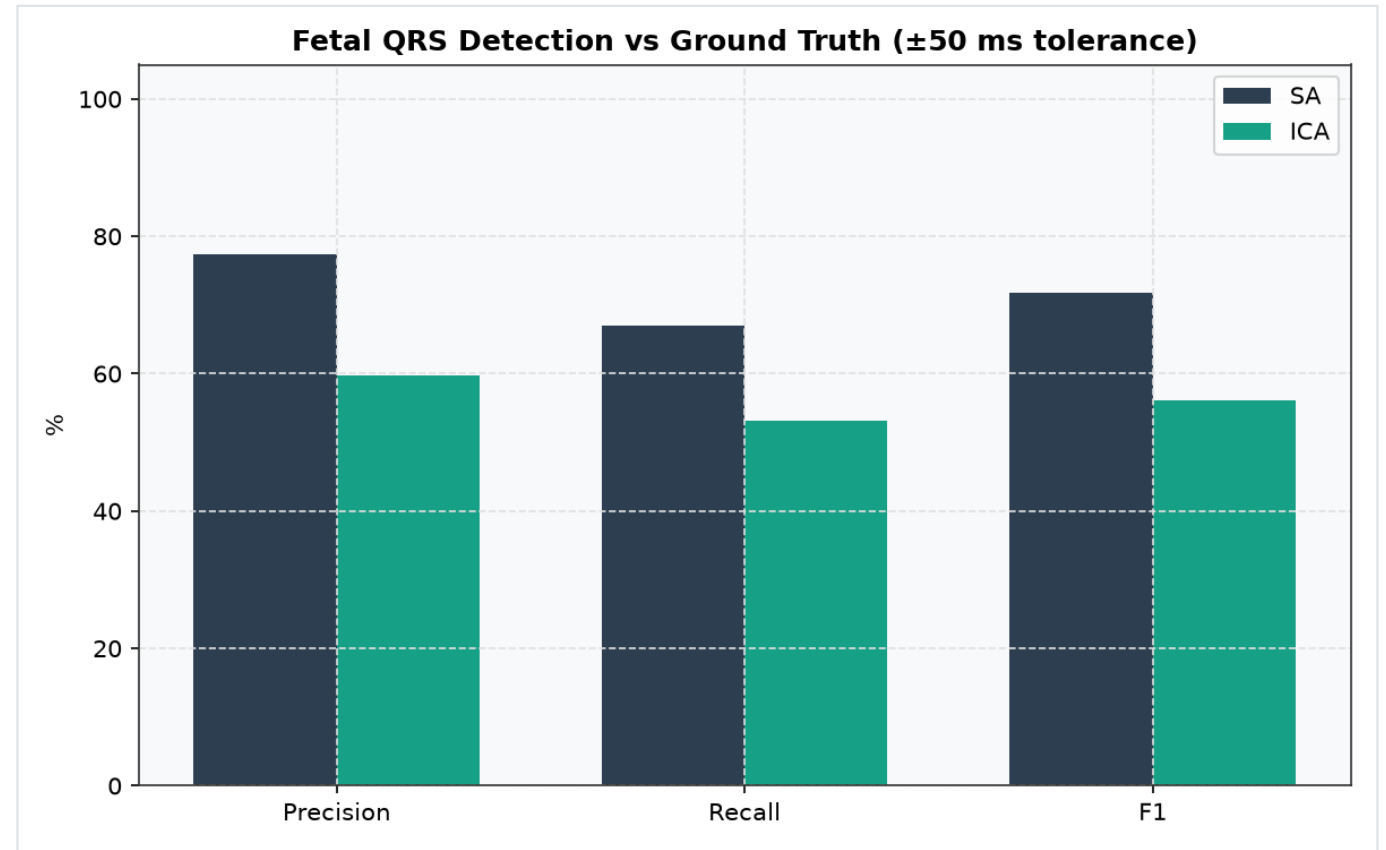


Figure 8. ground-truth precision / recall / F1

Results III: FHR value accuracy (MAE/RMSE)

- SA: MAE 13.5 bpm / RMSE 17.4 bpm
- ICA: MAE 18.0 bpm / RMSE 22.2 bpm

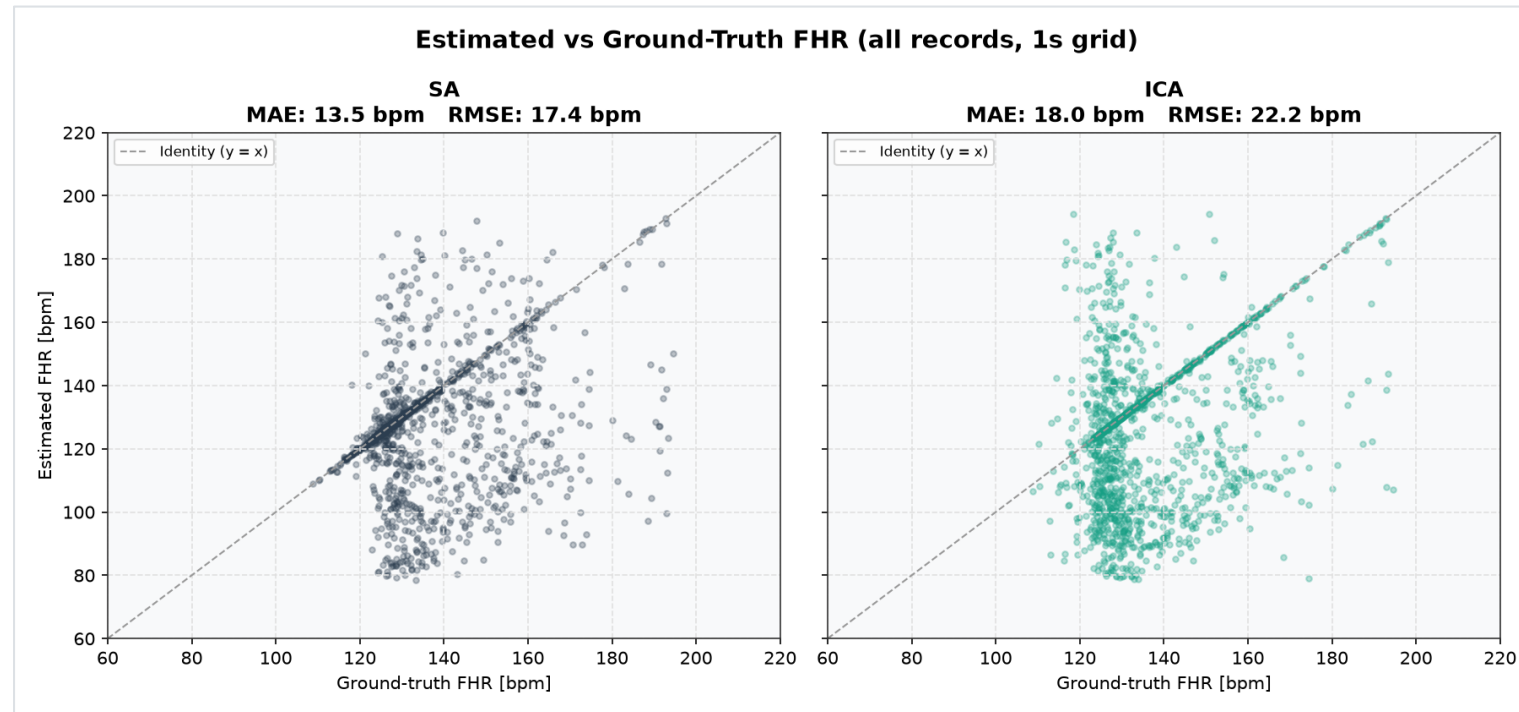


Figure 9. FHR-value accuracy

Results IV: robustness to matching tolerance

- F1 recomputed at 10 / 25 / 50 / 75 / 100 ms ground truth matching tolerances
- SA leads *ICA* at **every** tolerance tested

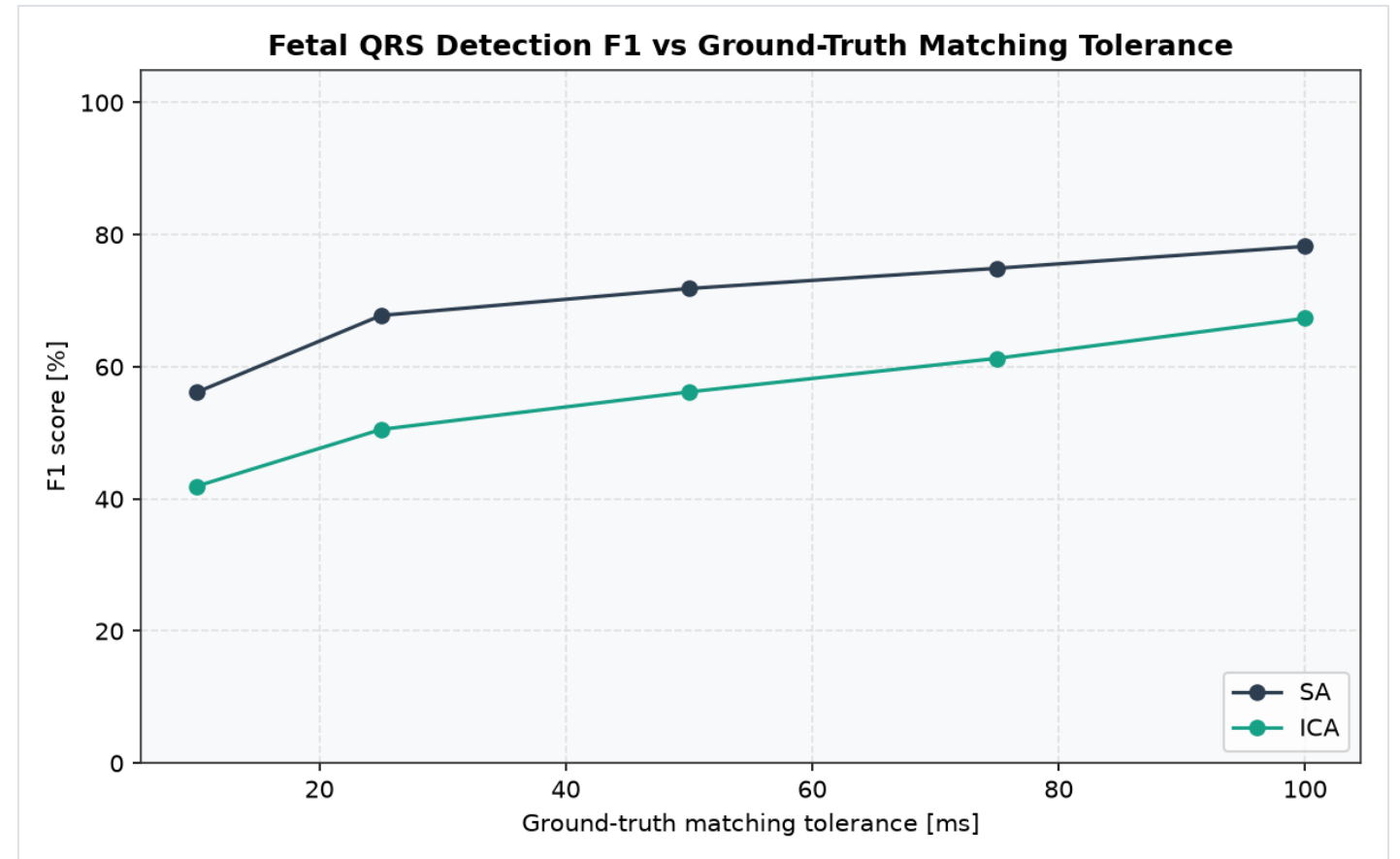


Figure 10. robustness to matching tolerance

Results V: signal quality drives reliability

- *Noise conditions*: mean $SNR \approx -4.9$ dB, mean $SIR \approx -12.0$ dB
- SA reliability degrades gracefully as SNR/SIR drop; ICA collapses more sharply
- *Spearman ρ* : SA– SNR **0.87** ($p < 0.001$), SA– SIR **0.50** ($p = 0.012$); ICA– SNR **0.10** ($p = 0.63$), ICA– SIR **0.18** ($p = 0.40$)

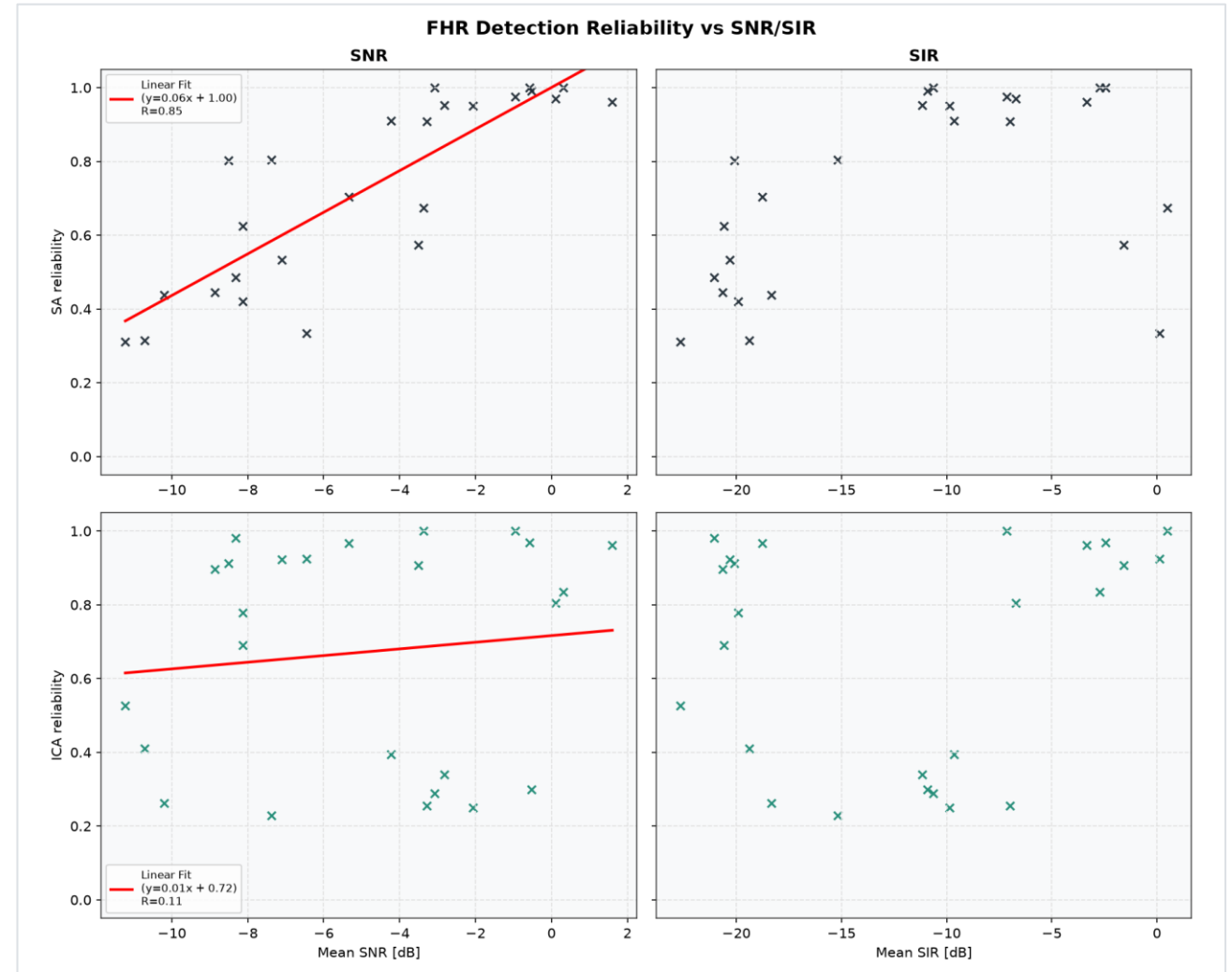
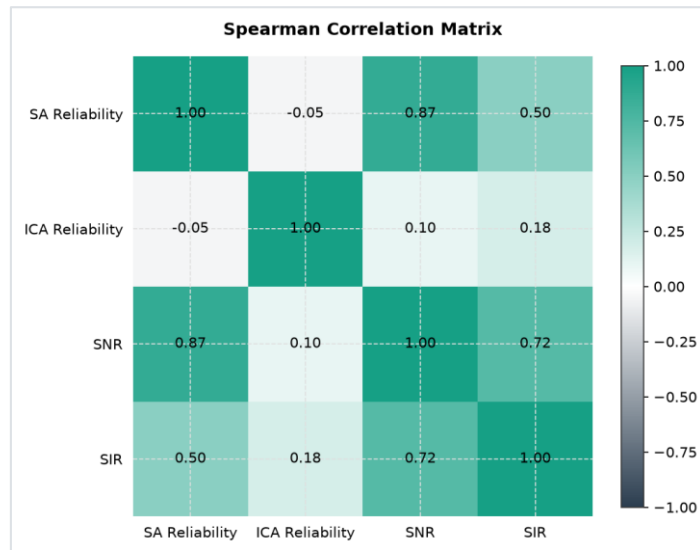


Figure 11. signal quality drives reliability

Discussion & Conclusions

- SA outperforms ICA by leveraging known signal structure yielding better accuracy in low *SNR* conditions,
- showing expected lower *recall* in specific records (a18, a09, a02, a11, a15) consistent with the paper's gestational age hypothesis,
- supporting the practical development of affordable and *non-invasive home-based* monitoring systems

Thank you